Convergence study of first order homogenization methods for steady-state heat conduction in periodically insulated media

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Abstract

We implement and validate a first-order periodic homogenization method for steady-state heat conduction in a plate perforated by a periodic array of insulating holes. The microstructure enters only through a pair of corrector problems on a single unit cell, which yield an effective conductivity tensor $\kappa^\star$; the macroscopic temperature T is then computed on a hole-free mesh and the fine-scale field is reconstructed pointwise from the two-scale expansion $T+\varepsilon\chi\cdot\nabla T$. The method is benchmarked against a direct numerical solve (DNS) on an exactly matching periodic tiling. With quadratic finite elements on curved meshes, and meshes refined until the results no longer change, the finite element error is driven well below the homogenization error. For a rotated elliptical hole, the RMS error of the macroscopic field alone decays as $O(\varepsilon^{0.97})$ in the cell size ε, while the error of the two-scale reconstruction decays as $O(\varepsilon^{2.02})$: adding the corrector raises the order of accuracy from one to two, as the asymptotic expansion predicts. The conductivity tensor measured on a finite perforated sample converges to $\kappa^\star$ at rate $O(\varepsilon)$, with no plateau, for both a circular and an elliptical hole. We also study how hole shape controls $\kappa^\star$: at constant porosity, elongating an ellipse raises conductivity along its major axis and lowers it across, while rotating it produces an off-diagonal entry that is largest near 45°.

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1 Introduction

The objective of this study is to verify the convergence of first order homogenization methods for heat conduction problems in periodically insulated media. Before homogenization is itself presented, motivation for its use is given. In particular, the naive approach of a Direct Numerical Solve (DNS) is discussed as the obvious method for computing the heat field on a perforated plate. This direct solve uses the Finite Element Method (FEM). Essential to this technique is the variational or weak form of the governing PDE – here the steady state heat equation – and as such a derivation of this equation is provided. Complementing the derivation is a brief explanation of how this weak form is used to build the solver for this study. While the DNS is viable and accurate – so much so that it is used as a benchmark throughout this investigation – for a small number of perforations, it quickly becomes too expensive once their number increases, for reasons discussed in Section 2.3.4. To overcome this, a first order homogenization method is introduced to solve a simpler problem and reconstruct the solution function without needing to produce complicated geometries or solve equations on them. The results of the this new, homogenized simulation are then presented and discussed before some brief concluding remarks are made to close the study.

2 DNS - Heat conduction and the Finite Element Method

This section provides the ingredients needed for the naive DNS. The heat equation, boundary conditions, weak form, and FEM are discussed.

2.1 The heat equation - governing physics

Consider a bounded domain $\Omega \subset \mathbb{R}^2$, with boundary $\partial\Omega$. The temperature $u(\boldsymbol{x}, t)$ obeys the general time-dependent heat equation

$$\begin{cases} \nabla \cdot (\kappa \nabla u) + Q = \rho c_p \frac{\partial u}{\partial t}, & \text{on } \Omega \\ \text{boundary condition}, & \text{on } \partial\Omega \\ u \equiv u_0, \quad t = 0 \end{cases} \tag{1}$$

where

- $\kappa(\boldsymbol{x})$ is the conductivity tensor
- ρ is the mass density of the medium
- c_p is the heat capacity at constant pressure
- Q is the total of all heat sources

In our study, the equation is drastically simplified so that emphasis may be placed on the numerical homogenization methods as opposed to sophisticated physics. In particular, the following assumptions are made:

1. There are no sources or sinks in the plate (i.e. $Q = 0$)
2. The plate has reached a steady-state temperature and (no time dependence, $\partial_t u = 0$)
3. Conductivity κ is constant and isotropic throughout the solid material (i.e. κ is a scalar everywhere it is defined)

Thus the problem transforms from an initial-boundary value problem to a boundary value problem. One might, of course, extend the analysis to the time-dependent case, whereby the physical quantities of interest could become the speed of heat propagation, transient storage, time constants, and thermal lag. This is an interesting problem in its own right, however the implementation is even more involved. With the assumptions above, Eq. (1) reduces to

$$\begin{cases} \nabla \cdot (\kappa \nabla u) = 0, & \text{on } \Omega \\ \text{boundary condition}, & \text{on } \partial\Omega \end{cases} \tag{2}$$

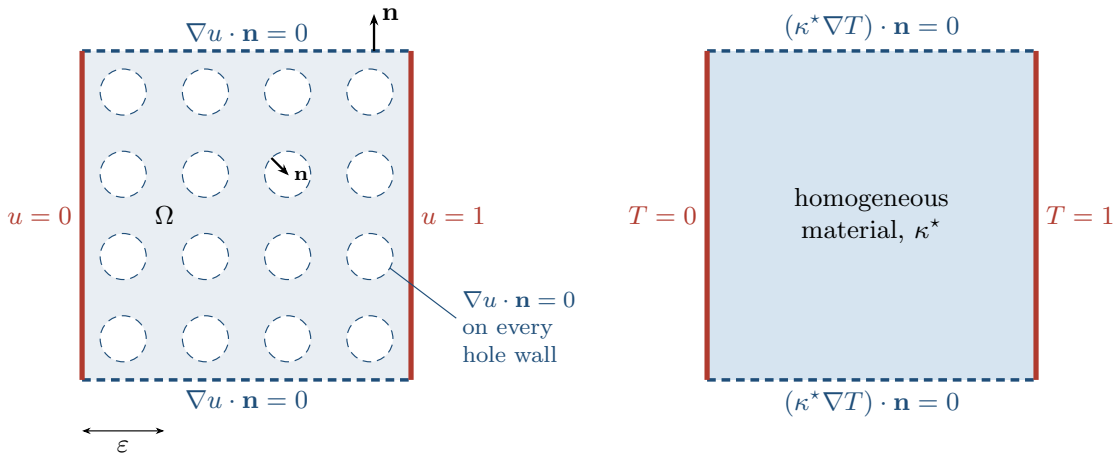
with $u = u(x)$ and $x = (x_1, x_2) \in \Omega$. This is the final, simplified equation which is used to govern the physics throughout the study.

2.2 Equation domain and boundary conditions

In order for the problem to be well posed, one must put into place the boundary conditions. The case of interest is when the domain has periodic inclusions made of an entirely insulating medium, that is, there are periodic occurrences of regions where $\kappa = 0$. In order to recreate this numerically a rectangular domain is used, which is then perforated with regularly spaced circular holes. Perfect insulation is simulated by imposing a Neumann boundary condition: through the surface of each such inclusion zero heat flux is allowed. Further, the quantity of interest in this study is the effective conductivity tensor $\kappa^\star$ of the resulting material. To gain information about this tensor, a macroscopic temperature gradient is imposed laterally on the whole domain. This is done by choosing Dirichlet boundary conditions. Finally, in order to avoid heat leakage through the bottom and top edges of the rectangle, zero heat flux is required through these edges also. A summary of these conditions, on the domain $\Omega = [0, L_1] \times [0, L_2]$, is given below

$$\text{BC} = \begin{cases} u(0, x_2) = 0 \quad \forall x_2 \in [0, L_2] \\ u(L_1, x_2) = 1 \quad \forall x_2 \in [0, L_2] \\ \nabla u \cdot \mathbf{n} = 0 \quad \text{on } y_{\min},\ y_{\max} \text{ edges} \\ \nabla u \cdot \mathbf{n} = 0 \quad \text{on inclusion edges} \end{cases} \tag{3}$$

A sketch of these conditions can be found in Fig. 1(a), and an example mesh, as used in the computations, is shown in Fig. 2.



(a) perforated plate, solved by DNS

(b) homogenized plate

Figure 1: Boundary conditions of the two problems posed on the plate. (a) The direct numerical solve of Eq. (2) with the conditions of Eq. (3), where the temperature is imposed on the left and right edges and no heat crosses the top and bottom edges or the wall of any hole, $\mathbf{n}$ being the outward normal of Ω, which points into the holes. (b) The homogenized problem of Eq. (12), posed on the same square without holes, with the same imposed temperatures and with insulated top and bottom edges, where the zero-flux condition now involves the effective tensor $\kappa^\star$.

One could extend the study to three dimensions, in which case the domain would resemble the one shown in Fig. 10 in the appendix. A slice of the direct numerical solve for a simple 10 by 10 case can be seen in Fig. 11. The cube is sliced to reveal the temperature inside it, which varies across the cylinders but not along them. Perforating a cube with simple parallel cylinders is, by symmetry, exactly equivalent to completing a simpler 2D study of the perpendicular cross-section, which is what is done here; reducing dimensions substantially reduces computational costs. It would be interesting, however, to study the effect of rotating or curving the inclusions on the effective conductivity tensor. One might ask if any $\kappa^\star$ can be attained by playing with the geometry.

2.3 Variational form

2.3.1 General presentation

Finding the variational or weak form of a partial differential equation is the first step in most finite element techniques. Not every PDE admits one: the construction below relies on integrating by parts,

which requires the equation to have a suitable structure. Whether a given problem has a weak form therefore has to be checked, and the next subsection does this for the heat equation considered here. Given a boundary value problem Ξ over a domain Ω, let V be the general solution space. Define further $\partial\Omega_{\text{Dirichlet}}$ and $\partial\Omega_{\text{Neumann}}$ as the parts of the boundary where Dirichlet or Neumann boundary conditions are applied, respectively. The variational form is obtained, in general, by multiplying both sides of the partial differential equation with a test function

$$v \in V^0 := \{f \in V : f(\mathbf{x}) = 0 \quad \forall \mathbf{x} \in \partial\Omega_{\text{Dirichlet}}\}$$

Then through integration by parts and integral identities, one obtains an equation of the form

$$a(u, v) = L(v) \quad \forall v \in V^0 \tag{4}$$

for some linear form $L : V \to \mathbb{R}$ and bilinear form $a : V \times V \to \mathbb{R}$, which must be satisfied by u. Once the problem is rewritten in this way, the infinite dimensional solution space V is approximated by a finite dimensional space V_h with a well chosen basis $\{v_h^i\}_{i=1}^n$ of functions. These functions are called "finite elements". In effect, the essence of the method consists in approximating the unknown function by assuming it is made up of a linear combination of these finite elements, and then determining the weight of each using Eq. (4). In a finite dimensional space, a bilinear form can be represented by its associated matrix $\mathbf{A}$, and a linear form by a vector of its coefficients in the dual space $\boldsymbol{l}$. In this way, Eq. (4) reduces to a linear problem

$$\mathbf{v}^\mathsf{T}\mathbf{A}\mathbf{u} = \mathbf{v}^\mathsf{T}\boldsymbol{l} \quad \forall \mathbf{v} \in V_h^0 \iff \mathbf{A}\mathbf{u} = \boldsymbol{l}$$

Further, the finite elements are generally chosen to be equal to 1 at their node on the mesh, and to decrease to zero everywhere else. As a result, the weights in front of each element are in fact equal to the value of the unknown function u at the corresponding node. A typical example would be the hat functions, or 1st order Lagrange elements. See https://jsdokken.com/FEniCS-workshop/README.html for a more comprehensive presentation of the finite elements.

2.3.2 Application

Consider again the boundary value problem in Eq. (2), with the boundary conditions of Eq. (3). Take $V = C^2(\bar{\Omega})$, the set of twice differentiable functions on the closure of Ω. We show that this problem admits a weak form, in three steps:

1. starting from the PDE, we derive an equation of the form Eq. (4);
2. we state the resulting weak form, for the boundary conditions used here and for more general Neumann data;
3. we show the converse: a function satisfying the weak form (and the Dirichlet conditions) solves the PDE and satisfies the Neumann conditions.

Steps 1 and 3 together show that solving the weak form is equivalent to solving the original problem, which is what justifies using it in the solver.

Throughout, $\mathbf{n}$ is the outward unit normal on $\partial\Omega$, dl the length element along the boundary, and $d\mathbf{l} = \mathbf{n}\, dl$ the corresponding normal vector element.

Step 1: from the PDE to the weak form. Let u solve Eq. (2), and multiply by any $v \in V^0$. Since κ is a non-zero constant,

$$\begin{aligned}
& \nabla \cdot (\kappa \nabla u) = 0 \\
\Rightarrow \quad & \Delta u\, v = 0 \\
\Rightarrow \quad & \iint_\Omega \Delta u\, v \, d\Omega = 0 \\
\Rightarrow \quad & \iint_\Omega \nabla \cdot (v \nabla u)\, d\Omega - \iint_\Omega \nabla u \cdot \nabla v \, d\Omega = 0 \\
\Rightarrow \quad & \oint_{\partial\Omega} v \nabla u \cdot d\mathbf{l} - \iint_\Omega \nabla u \cdot \nabla v \, d\Omega = 0 \qquad \forall v \in V^0
\end{aligned}$$

where the fourth line uses the product rule $\nabla \cdot (v\nabla u) = \nabla u \cdot \nabla v + v\Delta u$ and the last one the divergence theorem. Splitting the boundary integral,

$$\oint_{\partial\Omega} v\nabla u \cdot d\mathbf{l} = \int_{\partial\Omega_{\text{Dirichlet}}} v\nabla u \cdot d\mathbf{l} + \int_{\partial\Omega_{\text{Neumann}}} v\nabla u \cdot d\mathbf{l}.$$

The first term is null because $v = 0$ on $\partial\Omega_{\text{Dirichlet}}$ by definition. The second is null because the Neumann conditions give $\nabla u \cdot d\mathbf{l} = (\nabla u \cdot \mathbf{n})\, dl = 0$ on $\partial\Omega_{\text{Neumann}}$.

Step 2: the weak form. We are left with

$$\iint_\Omega \nabla u \cdot \nabla v \, d\Omega = 0 \quad \forall v \in V^0 \tag{5}$$

which is of the form of Eq. (4), with

$$a(u, v) = \iint_\Omega \nabla u \cdot \nabla v \, d\Omega\,, \quad L(v) = 0.$$

The Dirichlet conditions do not appear in Eq. (5): they are imposed on u directly, and through the choice $v \in V^0$. The Neumann conditions, on the other hand, are contained in the linear form L. With a general Neumann condition $\nabla u \cdot \mathbf{n} = g \in C^0(\partial\Omega_{\text{Neumann}})$, the second boundary term in Step 1 no longer vanishes but equals $\int_{\partial\Omega_{\text{Neumann}}} gv\, dl$, and the same derivation gives

$$\iint_\Omega \nabla u \cdot \nabla v \, d\Omega = \int_{\partial\Omega_{\text{Neumann}}} gv\, dl \quad \forall v \in V^0, \qquad \text{i.e.} \quad L(v) = \int_{\partial\Omega_{\text{Neumann}}} gv\, dl. \tag{6}$$

Our problem is the case $g = 0$.

Step 3: from the weak form back to the PDE. Conversely, let $u \in V$ satisfy the Dirichlet conditions and Eq. (6). Running the integrations by parts of Step 1 backwards,

$$\begin{aligned}
& \iint_\Omega \nabla u \cdot \nabla v \, d\Omega = \int_{\partial\Omega_{\text{Neumann}}} gv\, dl \\
\Rightarrow \quad & \iint_\Omega \nabla \cdot (v\nabla u)\, d\Omega - \iint_\Omega \Delta u\, v \, d\Omega = \int_{\partial\Omega_{\text{Neumann}}} gv\, dl \\
\Rightarrow \quad & \oint_{\partial\Omega} v\nabla u \cdot d\mathbf{l} - \iint_\Omega \Delta u\, v \, d\Omega = \int_{\partial\Omega_{\text{Neumann}}} gv\, dl \\
\Rightarrow \quad & \iint_\Omega \Delta u\, v \, d\Omega = \int_{\partial\Omega_{\text{Dirichlet}}} v\nabla u \cdot d\mathbf{l} + \int_{\partial\Omega_{\text{Neumann}}} v\nabla u \cdot d\mathbf{l} - \int_{\partial\Omega_{\text{Neumann}}} gv\, dl
\end{aligned}$$

and, since $v = 0$ on $\partial\Omega_{\text{Dirichlet}}$,

$$\iint_\Omega \Delta u\, v \, d\Omega = \int_{\partial\Omega_{\text{Neumann}}} v\,(\nabla u \cdot d\mathbf{l} - g\, dl) = \int_{\partial\Omega_{\text{Neumann}}} v\,(\nabla u \cdot \mathbf{n} - g)\, dl \quad \forall v \in V^0. \tag{7}$$

Equation (7) holds for every test function, and choosing the test function cleverly recovers, in turn, the PDE and the Neumann conditions.

The PDE. Restrict Eq. (7) to test functions that vanish on the whole boundary. The right-hand side is then zero, so $\iint_\Omega \Delta u\, v\, d\Omega = 0$ for all such v. Suppose that $\Delta u(\mathbf{x}_0) > 0$ at some interior point $\mathbf{x}_0$. Since $u \in C^2$, Δu is continuous, so it stays positive on a small disc around $\mathbf{x}_0$; taking for v a smooth bump that is positive inside this disc and zero outside gives $\iint_\Omega \Delta u\, v\, d\Omega > 0$, a contradiction (the case $\Delta u(\mathbf{x}_0) < 0$ is identical). Hence $\Delta u = 0$ in Ω, i.e. $\nabla \cdot (\kappa\nabla u) = 0$.

The Neumann conditions. Now that $\Delta u = 0$, the left-hand side of Eq. (7) vanishes for every $v \in V^0$, leaving

$$\int_{\partial\Omega_{\text{Neumann}}} v\,(\nabla u \cdot \mathbf{n} - g)\, dl = 0 \quad \forall v \in V^0.$$

The same argument, now with bumps centred on points of $\partial\Omega_{\text{Neumann}}$ (which may be non-zero there, since only the Dirichlet part of the boundary is excluded), shows that $\nabla u \cdot \mathbf{n} - g = 0$ on $\partial\Omega_{\text{Neumann}}$: the length element dl drops out and the Neumann condition is recovered.

Hence the PDE and the Neumann conditions are both contained in the weak form. This is why, when implementing the solver, the only boundary conditions one needs to specify are the Dirichlet conditions, and these fix the uniqueness of the solution.

2.3.3 Discretization: quadratic elements on curved meshes

All computations in this study use second-order Lagrange elements, called P2, on second-order, curved triangular meshes, rather than the hat functions above. The reason is the size of the quantity we want to measure. The homogenization error of the two-scale reconstruction turns out to be around 2×10^{-5} at the smallest cell size considered, and it can only be measured if the finite element error is well below that. This is harder than it sounds, because the DNS mesh must resolve every hole. Its element size is tied to the cell size, $h \propto \varepsilon$, so its relative accuracy does not improve as $\varepsilon \to 0$, and any quantity whose homogenization error is smaller than this fixed FEM error simply plateaus. This is what happened in an earlier version of this study with first-order elements, where for a circular hole the effective conductivity measured from the DNS differed from $\kappa^\star_{11}$ by a constant 0.6% at every ε. For a circle with the boundary conditions of Eq. (3) the homogenized solution is exact, as explained in Section 3.2.3, so this entire 0.6% was discretization error, and with P2 elements on curved meshes of the same element size it falls to 5×10^{-5}.

Curved elements matter as much as the polynomial degree. A hole meshed with straight-edged triangles is a polygon, and the resulting geometric error, $O(h^2)$, caps the accuracy whatever the degree of the elements. Second-order meshes place the mid-edge nodes on the hole boundary, which reduces the geometric error to $O(h^3)$ and lets the P2 elements reach their nominal accuracy. The computed $\kappa^\star$, an energy quantity, then converges like h^4, and changes by only 3×10^{-7} between unit-cell meshes of element size 1.5/120 and 1.5/160. Figure 2 shows a DNS mesh exactly as it is used in Section 3.2.3, at the finest of the three resolutions, together with a close-up of one hole on which the curved edges and the mid-edge nodes of the elements can be seen. The three-dimensional mesh of Fig. 10, which only illustrates a possible extension of the study, was made with straight-edged first-order elements.

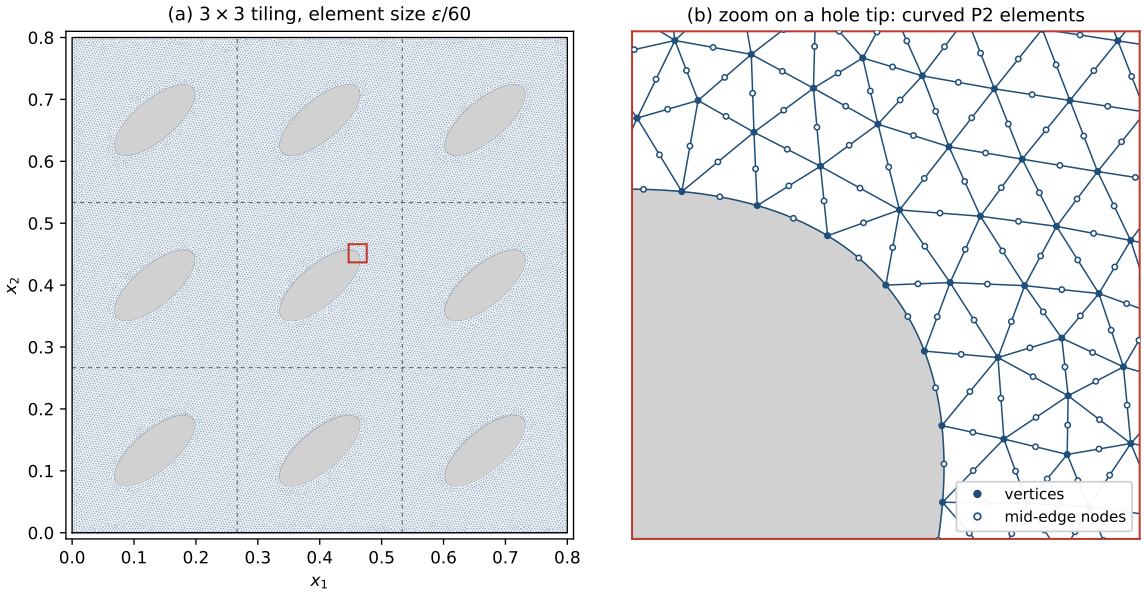


Figure 2: DNS mesh exactly as used in Section 3.2.3, for 3×3 cells of the rotated elliptical hole ($a = 0.3$, $b = 0.12$, 40°) on $[0, 0.8]^2$, with second-order triangles of size $\varepsilon/60$, the finest of the three mesh levels of the convergence study. (a) The whole plate, with the cell boundaries dashed; at this resolution the individual elements are barely visible. (b) Close-up of the region framed in (a), at the tip of a hole, where each edge of the six-node triangles is drawn as the quadratic curve through its mid-edge node, so that the edges lying on the hole follow its boundary instead of cutting across it. Generated by `scripts/06_mesh_figure.py`.

2.3.4 DNS - discussion

The main difficulty of the direct numerical solve comes from the holes themselves. Since each hole must be resolved individually, the mesh has to be refined around every inclusion, and the total number of elements grows roughly in proportion to the number of holes for a fixed resolution per hole. The resolution needed to resolve the effects of many small inclusions therefore becomes too expensive, and beyond a modest number of perforations the linear system is so large that the simulation fails to terminate within a

reasonable time frame. The computational domain to mesh also becomes much more complicated, to the extent that even building the geometry, which means constructing and meshing hundreds of small cutouts in `gmsh`, becomes slow and fragile, independently of the linear solve. These two difficulties, the complexity of the geometry and the size of the system, are precisely what motivate the homogenization approach introduced next, in which the microstructure is replaced by an equivalent smooth material. This way, the fine geometry only ever needs to be resolved once on a single small unit cell, rather than once per macroscopic problem.

3 Homogenization

The DNS approach discussed above solves the exact problem at the cost of resolving every inclusion explicitly. Homogenization instead tries to solve the following problem: viewed from far enough away that individual holes are no longer visible, what effective, hole-free material would produce the same macroscopic temperature field? If such an equivalent material can be found, the macroscopic problem can be solved on a plain, coarse mesh with no microstructure to resolve at all, at a fraction of the DNS cost. This section develops that idea: first defining the effective conductivity tensor that characterizes the equivalent material, then deriving how it is computed from a single small unit-cell problem, and finally showing how the fine-scale temperature field can be reconstructed from the coarse solution without ever meshing the real, perforated domain.

3.1 Effective conductivity tensor

The effective conductivity tensor $\kappa^\star$ is the conductivity of the fictitious, homogeneous material that reproduces the same macroscopic behaviour as the true, perforated medium, once microscopic fluctuations have been averaged out. Formally, if $T(x)$ denotes the macroscopic temperature field (the local cell-average of the true, fine-scale temperature u) and q the macroscopic heat flux, the two are related by the constitutive law

$$q = -\kappa^\star \nabla T \tag{8}$$

A tensor, rather than a single scalar, is required in general because a perforated medium need not conduct equally well in every direction: an elongated or rotated inclusion can make the material conduct more readily along one axis than another, and can even deflect flux sideways relative to the imposed gradient. The diagonal entries $\kappa^\star_{11}, \kappa^\star_{22}$ describe conduction in directions 1 and 2 respectively; the off-diagonal entries $\kappa^\star_{12} = \kappa^\star_{21}$ (symmetry is discussed below) describe this cross-direction coupling, and vanish whenever the microstructure possesses the appropriate mirror symmetries (e.g. a circular or axis-aligned elliptical inclusion). We are interested in $\kappa^\star$ both because it is the object that makes the macroscopic problem solvable without resolving the microstructure, and because it provides a compact, coordinate-free summary of how a given inclusion shape affects heat transport.

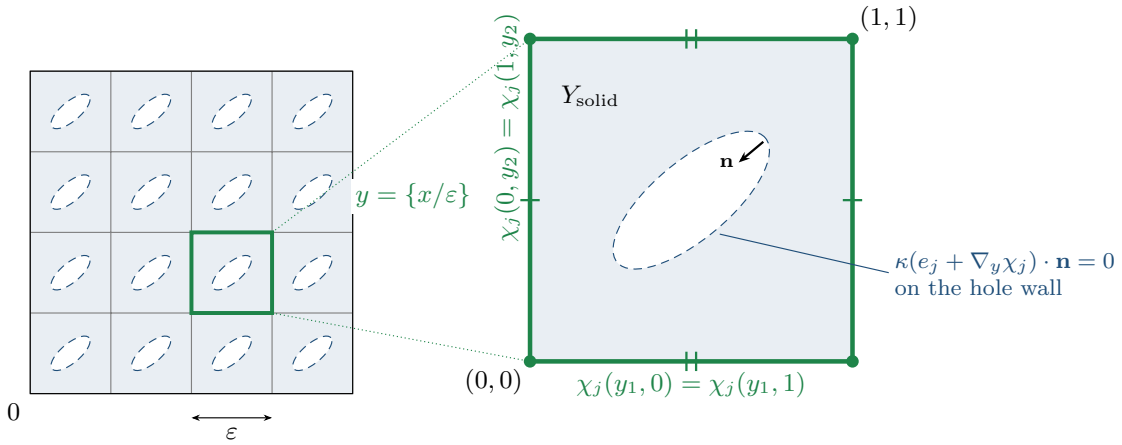
3.1.1 Computing the effective conductivity

Two length scales coexist in the perforated plate, and the notation keeps them apart. Positions in the plate are given by the macroscopic coordinate $x = (x_1, x_2) \in \Omega$. If ε is the side length of one periodic cell, rescaling a single cell and its inclusion to unit size gives the unit cell $Y = [0,1]^2$, in which positions are given by the microscopic coordinate $y = (y_1, y_2) \in Y$. A point x of the plate thus sits at the position $y = \{x/\varepsilon\}$ within its own cell, where $\{\cdot\}$ denotes the fractional part taken componentwise. We write Y_{solid} for the unit cell with the inclusion removed, ∇_y for the gradient with respect to y, and e_1, e_2 for the unit vectors of the two coordinate directions. The two coordinates and the correspondence between them are sketched in Fig. 3(a).

The effective tensor is obtained from two auxiliary problems posed on the unit cell alone, one for each direction. For $j \in \{1,2\}$, the corrector χ_j is the function on Y_{solid} that solves

$$\begin{cases} \nabla_y \cdot (\kappa\,(e_j + \nabla_y \chi_j)) = 0 & \text{in } Y_{\text{solid}} \\ \kappa\,(e_j + \nabla_y \chi_j) \cdot \mathbf{n} = 0 & \text{on the inclusion wall} \\ \chi_j(0, y_2) = \chi_j(1, y_2), \quad \chi_j(y_1, 0) = \chi_j(y_1, 1) & \forall y_1, y_2 \in [0,1] \\ \displaystyle\int_{Y_{\text{solid}}} \chi_j \, dy = 0 \end{cases} \tag{9}$$

The third condition makes χ_j periodic on the cell. The first three conditions determine χ_j only up to an additive constant, since adding a constant changes neither its gradient nor its periodicity, and the zero-mean condition removes this freedom so that the solution is unique. It is also the natural normalization, consistent with T being the average of the fine-scale temperature over a cell. Figure 3(b) shows on which part of the cell boundary each of the other conditions applies.



(a) plate Ω, coordinate x (b) unit cell Y, coordinate y

Figure 3: The two coordinates and the cell problem. (a) The plate is a tiling of cells of side ε, and the fractional part $y = \{x/\varepsilon\}$ maps every one of them onto the same unit cell. (b) The boundary conditions of the cell problem of Eq. (9). The corrector takes equal values on the left and right faces and on the bottom and top faces, the faces being paired by the single and double tick marks, while the hole wall is insulated. The four corners, marked by dots, are all tied to $(0, 0)$ in the discrete problem.

The meaning of the correctors is clearest through the functions

$$t_j(y) = y_j + \chi_j(y), \qquad j \in \{1, 2\}.$$

Since $\nabla_y t_j = e_j + \nabla_y \chi_j$, Eq. (9) states that t_j is a steady temperature field in the solid part of the cell, insulated on the inclusion wall, whose values on opposite faces differ by exactly one in direction j and coincide in the other direction. In other words, t_j is the temperature of a cell subjected to a unit macroscopic gradient e_j. Without the inclusion it would be the linear profile y_j, and the corrector χ_j is precisely the small periodic deviation from this profile that the inclusion forces as heat is diverted around it, vanishing identically when there is no hole.

Multiplying by a periodic test function and integrating by parts as in Section 2.3.2 gives the weak form solved in practice, in which the boundary terms vanish because the contributions of opposite faces cancel by periodicity and no heat crosses the inclusion wall. It reads: find χ_j, periodic on Y with zero mean, such that

$$\int_{Y_{\text{solid}}} \kappa \nabla_y \chi_j \cdot \nabla_y v \, dy = -\int_{Y_{\text{solid}}} \kappa e_j \cdot \nabla_y v \, dy \quad \forall v \; Y\text{-periodic}\,. \tag{10}$$

The effective tensor is then the average, over the whole cell, of the local heat flux produced by a unit gradient in direction j,

$$\kappa^\star_{ij} = \frac{1}{|Y|} \int_{Y_{\text{solid}}} \kappa \, (e_j + \nabla_y \chi_j) \cdot e_i \, dy = \frac{1}{|Y|} \int_{Y_{\text{solid}}} \kappa \frac{\partial t_j}{\partial y_i} \, dy, \tag{11}$$

so that its j-th column is the mean flux carried by a cell subjected to the gradient e_j. The normalization uses the area $|Y|$ of the full cell, inclusion included, rather than the solid area alone, and this is what accounts for the loss of conductivity due to porosity. Taking $v = \chi_i$ in Eq. (10) shows that

$$\int_{Y_{\text{solid}}} \kappa (e_j + \nabla_y \chi_j) \cdot \nabla_y \chi_i \, dy = 0$$

which turns Eq. (11) into

$$\kappa^\star_{ij} = \frac{1}{|Y|}\int_{Y_{\text{solid}}} \kappa \nabla_y t_i \cdot \nabla_y t_j \, dy.$$

This form is manifestly symmetric in i and j, so $\kappa^\star$ is symmetric whenever the local conductivity κ is, which here, κ being a scalar, is always the case. The property is used throughout this study as a check on the implementation.

In practice the two cell problems are solved once per inclusion shape, on a mesh of the unit cell only, which can therefore afford a high resolution. Periodicity is enforced both on the mesh, whose nodes are placed identically on opposite faces, and on the linear system, in which each degree of freedom on the right face is tied to its image on the left face and each on the top face to its image on the bottom. The three corners $(1,0)$, $(0,1)$ and $(1,1)$, marked in Fig. 3(b), are tied directly to $(0,0)$, since the corner $(1,1)$ would otherwise be constrained twice and the computed corrector could jump across the faces near it. The zero-mean condition is imposed in two stages, by pinning one degree of freedom to remove the additive constant during the solve and shifting the result to zero mean afterwards. The resulting system is small and is solved with a direct LU solver, which gives $\kappa^\star_{12} = \kappa^\star_{21}$ to machine precision.

3.1.2 Effective conductivity and perforation shapes

Figure 4 shows the temperature field obtained from a single unit cell problem, together with the local heat flux, for a circular and an elliptical inclusion, and Figs. 12 and 13, in the appendix, show the corresponding fields over a whole perforated plate. In both cases the flux visibly bends around the inclusion, as the zero-flux condition of Eq. (9) requires, since no heat can cross into the hole and the flux must therefore remain tangent to its boundary. For the circular inclusion the resulting tensor is isotropic and diagonal to within numerical precision, with $\kappa^\star_{11} = \kappa^\star_{22}$ and $\kappa^\star_{12} = 0$, as expected from the four-fold rotational symmetry of a circle in a square cell. No direction is geometrically preferred, so none can be preferred by the effective conductivity either.

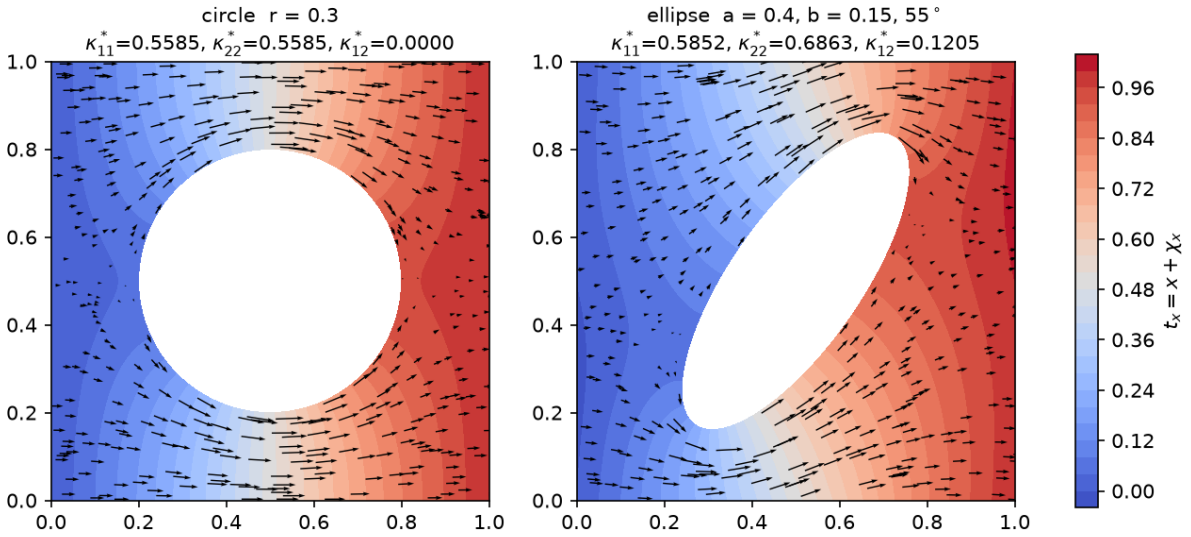
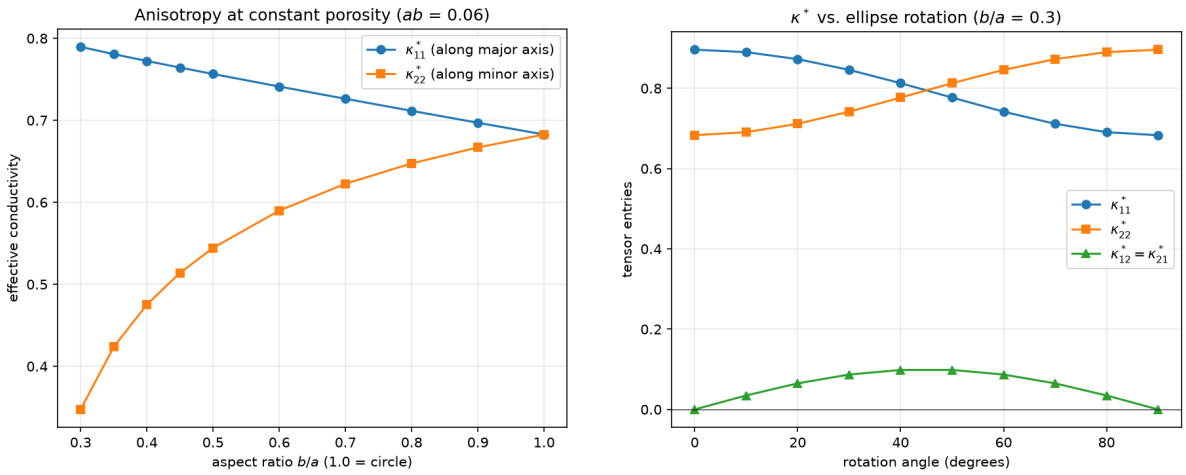


Figure 4: Unit-cell solution for a unit gradient in direction 1: local temperature $t_1 = y_1 + \chi_1$ (colour) and heat flux (arrows) for the circular and elliptical inclusions. The flux is tangent to the hole wall and squeezes through the narrow gaps; for the tilted ellipse it is deflected off-axis, which is what the non-zero $\kappa^\star_{12}$ records. Generated by `scripts/03_cell_fields.py`.

Elongating the inclusion into an ellipse breaks this symmetry and makes the anisotropy directly visible. Done naively, elongation changes two things at once, because at fixed semi-axis a a thinner ellipse is also a smaller hole, which lowers the porosity and lets every direction conduct better. To isolate the effect of shape, Fig. 5(a) varies the aspect ratio b/a of an axis-aligned ellipse at constant hole area $ab = 0.06$, that is, at constant porosity. The expected trade-off then appears. Heat is increasingly free to travel along the elongated axis, so that $\kappa^\star_{11}$ rises, and increasingly obstructed across it, so that $\kappa^\star_{22}$ falls, steeply so once the gap between neighbouring holes becomes narrow. The trade-off is far from symmetric, as over the range shown $\kappa^\star_{22}$ loses about three times as much as $\kappa^\star_{11}$ gains.



(a) (b)

Figure 5: Effective conductivity tensor of an elliptical hole as its shape changes. (a) Entries $\kappa^\star_{11}$ and $\kappa^\star_{22}$ of an axis-aligned ellipse as a function of its aspect ratio b/a, at constant hole area $ab = 0.06$ and hence constant porosity, $b/a = 1$ being a circle. $\kappa^\star_{11}$ rises and $\kappa^\star_{22}$ falls as the ellipse is elongated in direction 1. (b) The three independent entries as a function of the rotation angle of the ellipse, at the fixed aspect ratio $b/a = 0.3$ with $a = 0.3$. Rotating the ellipse away from the coordinate axes generates a non-zero off-diagonal entry, as the imposed gradient is increasingly deflected off-axis by the tilted inclusion. Generated by `scripts/02_ellipse_sweeps.py`.

An axis-aligned ellipse still possesses mirror symmetry about both coordinate axes, however, so $\kappa^\star_{12}$ remains zero whatever its aspect ratio, and breaking this remaining symmetry requires rotating the ellipse. Figure 5(b) follows the three independent entries of $\kappa^\star$ as a fixed, elongated ellipse is rotated from 0° to 90°. The off-diagonal entry vanishes at the two axis-aligned extremes and is largest near 45°, where the principal axis of the inclusion is furthest from both coordinate directions. Physically, this is the tensor's way of recording that a macroscopic gradient imposed purely in direction 1 now produces a flux with a genuine component in direction 2, the tilted inclusion deflecting heat sideways. The sweep also reflects the symmetry of the square cell, since rotations by θ and by $90° - \theta$ give mirror-image holes, for which $\kappa^\star_{11}$ and $\kappa^\star_{22}$ swap while $\kappa^\star_{12}$ is unchanged. Throughout, the measured asymmetry $|\kappa^\star_{12} - \kappa^\star_{21}|$ stays at the level of machine precision, around 10^{-15}, as the symmetry argument above requires.

3.2 First order approximation

Section 3.1 obtained the effective tensor $\kappa^\star$ from a single, cheap unit-cell computation. This subsection uses it to solve the macroscopic problem without ever meshing the perforated domain, and then recovers the fine-scale temperature from the coarse solution.

3.2.1 Macroscopic problem and fine-scale reconstruction

With $\kappa^\star$ in hand, the macroscopic temperature T solves the homogeneous, hole-free problem

$$\begin{cases} \nabla \cdot (\kappa^\star \nabla T) = 0, \quad \text{on } \Omega \\ \text{same Dirichlet / Neumann boundary conditions as Eq. (3)} \end{cases} \tag{12}$$

on the macroscopic domain Ω, meshed as a plain rectangle without a single inclusion and sketched in Fig. 1(b). On the insulated top and bottom edges the zero-flux condition now reads $(\kappa^\star \nabla T) \cdot \mathbf{n} = 0$, the flux being given by the constitutive law of Eq. (8). This is the DNS problem of Eq. (2) with the scalar conductivity κ replaced by the constant and generally anisotropic tensor $\kappa^\star$, and since nothing small remains to be resolved, a coarse mesh suffices.

The fine-scale temperature is then expressed in terms of T and the correctors by the two-scale expansion

$$u^\varepsilon(x) = T(x) + \varepsilon \sum_{j=1}^{2} \chi_j \left(\left\{ \frac{x}{\varepsilon} \right\} \right) \frac{\partial T}{\partial x_j}(x) + O(\varepsilon^2), \tag{13}$$

whose first two terms we abbreviate as $T + \varepsilon\chi \cdot \nabla T$, with $\chi = (\chi_1, \chi_2)$. The remainder is the theoretical error of the method. Away from the boundary of Ω, near which the expansion breaks down for reasons discussed below, homogenization theory predicts that the first neglected term is of order ε^2, so that the reconstruction $T + \varepsilon\chi \cdot \nabla T$ approximates the fine-scale temperature to second order in ε, whereas T alone, which ignores the oscillation that the holes imprint on the temperature, is only accurate to first order. Verifying these two rates is the main purpose of Section 3.2.5.

Evaluating the reconstruction at a point x requires T and its gradient at x, read off the coarse macroscopic solution, and the correctors at the position $y = \{x/\varepsilon\}$ of x within its cell. Because the correctors are periodic and were computed once on the unit cell, this single small computation stands in for every cell of the plate, and no mesh of the full microstructure is ever built. The reconstruction is an explicit formula rather than a further PDE, so it can be evaluated at as many points as desired, along a line, on a grid for visualization or at the sampling points of a DNS comparison, at negligible cost.

3.2.2 Algorithm

The full homogenization pipeline implemented in this study consists of the following steps.

1. **Cell problem.** Mesh the unit cell Y with the desired inclusion, registering periodicity on opposite faces, and solve the two cell problems of Eq. (10), one per direction, normalizing each corrector to zero mean.

2. **Effective tensor.** Assemble the 2×2 tensor $\kappa^\star$ from the two correctors via Eq. (11).

3. **Macroscopic solve.** Mesh the macroscopic domain Ω coarsely, without any microstructure, and solve Eq. (12) for T with the constant tensor obtained in step 2.

4. **Reconstruction.** Wherever the fine-scale temperature is needed, evaluate $T + \varepsilon\chi \cdot \nabla T$ using T and its gradient from step 3 and the correctors from step 1.

The gain over the DNS of Section 2.3.4 comes from a clean separation of scales. Step 1 is the only step in which the microstructure is meshed, and it is solved once, on a mesh containing a single inclusion, however large or complex the macroscopic domain may be. Step 3 is a coarse solve on a hole-free mesh and step 4 is the pointwise evaluation of an explicit formula, so the cost of both is essentially independent of ε. Packing more and smaller cells into the same domain leaves the macroscopic mesh unchanged, whereas the cost of the DNS grows sharply with the number of inclusions it must resolve. The one-time cost of step 1 is amortized further whenever $\kappa^\star$ is reused for other domain shapes, boundary conditions or load cases, none of which require solving the cell problem again.

3.2.3 Validation setup

The homogenization pipeline was validated against a DNS on a geometry built so that the two are directly comparable. The DNS domain is an exact tiling of the unit cell used in the corrector problems, with the same hole shape and size relative to the cell and the cells arranged edge to edge without margins, so that the DNS and the homogenized model describe, by construction, the same material. The domain is the square $[0, L]^2$ with $L = 0.8$, divided into $n \times n$ cells of size $\varepsilon = L/n$ with n from 2 to 12, and every solve, whether of the cell problem, the macroscopic problem or the DNS, uses P2 elements on curved meshes as described in Section 2.3.3.

Choice of hole. A circular hole cannot show the homogenization error with the boundary conditions of Eq. (3). For a centred circle, $\kappa^\star$ is isotropic, so the macroscopic solution is the linear profile $T = x_1/L$, and the mirror symmetries of the cell make χ_1 vanish on the cell faces $y_1 = 0, 1$ and $\partial\chi_1/\partial y_2$ vanish on $y_2 = 0, 1$. The reconstruction $x_1/L + (\varepsilon/L)\,\chi_1(\{x/\varepsilon\})$ then satisfies the heat equation, the insulated hole walls, both Dirichlet conditions and the insulated top and bottom edges. It is thus the exact solution of the DNS problem, and any measured difference is FEM error. We therefore use a rotated ellipse, $a = 0.3$, $b = 0.12$, rotated by $40°$, for which

$$\kappa^\star \approx \begin{pmatrix} 0.78 & 0.09 \\ 0.09 & 0.75 \end{pmatrix}.$$

Because $\kappa^\star_{12} \neq 0$, the insulated top and bottom edges now require $\kappa^\star_{21}\partial T/\partial x_1 + \kappa^\star_{22}\partial T/\partial x_2 = 0$, so T is no longer linear and the two-scale expansion has a genuine error.

3.2.4 Boundary layers

The two-scale expansion of Eq. (13) is built from correctors that only know about the periodic cell, and nothing in its construction refers to the boundary of Ω. Away from the boundary this does no harm, but on $\partial\Omega$ the reconstruction does not in general satisfy the boundary conditions, and the true solution has to make up the difference within a thin region along the boundary, which we call (rather unimaginatively) a boundary layer. This subsection introduces the finite-sample problem in which the layer is easiest to see, explains why the layer appears, measures how thick it is, and uses this measurement to predict how the conductivity measured on a finite sample converges to $\kappa^\star$.

A finite sample. Besides the field comparison based on Eq. (3), this study measures the conductivity tensor of a finite perforated sample and compares it with $\kappa^\star$. The sample is the $n\times n$ tiling of Section 3.2.3, on which the DNS is solved twice, with the affine temperature $u_j = x_j$ imposed on the whole outer boundary for $j = 1, 2$ and the hole walls insulated, as sketched in Fig. 6. The apparent tensor of the sample is its volume-averaged flux,

$$\kappa_{ij}^{\text{app}} = \frac{1}{|\Omega|}\int_{\Omega_{\text{solid}}} \kappa\,\partial_i u_j\;dx, \tag{14}$$

normalized, like $\kappa^\star$, by the full area $|\Omega|$. The boundary layer will affect this quantity directly, because the flux is integrated over the whole sample, including the layer.

Why a layer appears. With the affine data $u = x_j$, the macroscopic solution is exactly $T = x_j$. The reconstruction $x_j + \varepsilon\chi_j(\{x/\varepsilon\})$ has the local gradient $e_j + \nabla_y\chi_j$, and by Eq. (9) it therefore satisfies the heat equation in the solid and the insulation condition on every hole wall exactly. Its only defect lies on the outer boundary, where it exceeds the imposed data by $\varepsilon\chi_j(\{x/\varepsilon\})$. Writing the true solution as

$$u^\varepsilon(x) = x_j + \varepsilon\chi_j\left(\left\{\frac{x}{\varepsilon}\right\}\right) + \varepsilon\theta(x), \tag{15}$$

the correction θ solves the same perforated problem as u^ε, with the boundary data $\theta = -\chi_j(\{x/\varepsilon\})$ on $\partial\Omega$. Since this data oscillates on the scale of a cell, one expects the correction to be felt only near the boundary, and the question is how far into the plate it reaches.

Measuring the layer. This question is answered numerically in Fig. 7, for the rotated elliptical hole of Section 3.2.3 and $u = x_1$ imposed on the whole outer boundary. Because the expansion is exact away from the boundary for this boundary data, the difference between the DNS and $x_1 + \varepsilon\chi_1$ is the correction $\varepsilon\theta$ alone, and its map in Fig. 7(a) is indeed confined to a narrow band along the boundary. To see how thick this band is, the difference is averaged over thin strips parallel to the boundary and plotted against the distance d to the boundary, measured in cells, that is d/ε, which gives the profiles of Fig. 7(b). Two observations stand out.

First, the profiles obtained for $n = 4$, 8 and 12 fall on a single curve. Measured in cells, the layer therefore looks the same whatever the cell size, which means that θ depends on x only through x/ε, and in physical units its thickness is proportional to ε. The opposite edges lie 4 cells away in the first case and 12 in the last, so the agreement also shows that each edge produces its own layer, unaffected by the others.

Second, this common curve is a straight line on the logarithmic scale, so the layer decays exponentially with the distance to the boundary,

$$\varepsilon\,|\theta| \propto \varepsilon\,e^{-\gamma d/\varepsilon}, \tag{16}$$

with a fitted rate $\gamma = 5.65$. The layer thus loses a factor ten every 0.41ε, and a little under one cell from the boundary it has fallen by more than two orders of magnitude and disappears below the discretization error. Further from the boundary nothing is left of it, the mean of θ beyond one and a half cells being of order 10^{-8}.

The same mechanism is at work in the problem of Eq. (3), where on the Dirichlet edges the reconstruction misses the imposed temperature by $\varepsilon\chi\cdot\nabla T$, and on the insulated edges the flux of the reconstruction vanishes only on average along each cell. This is why the field errors are best measured at a distance from the edges.

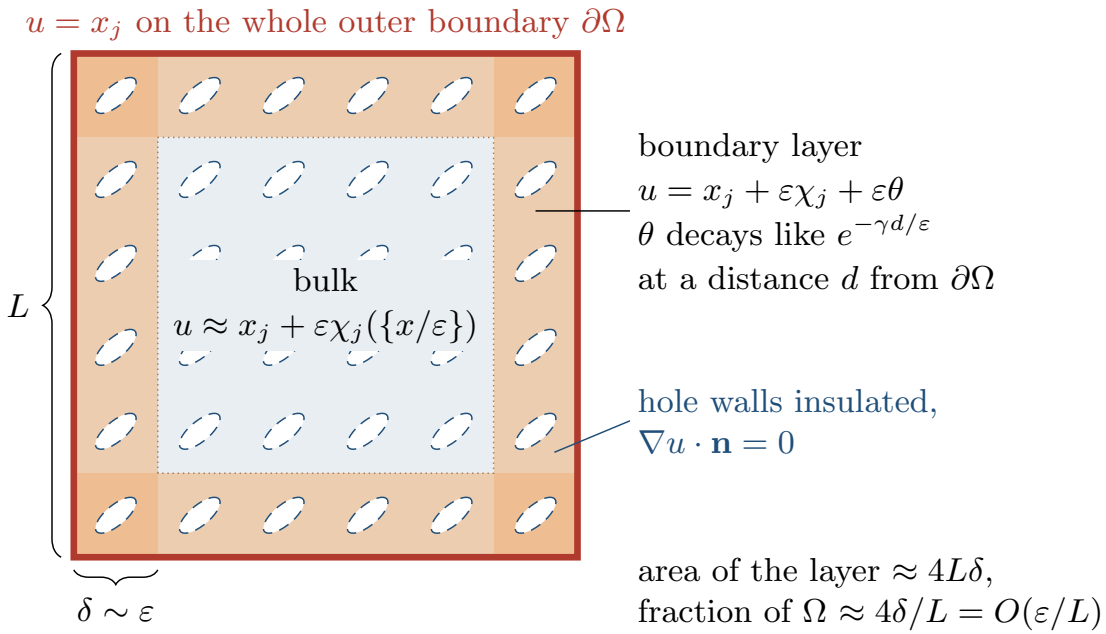


Figure 6: Boundary layer of a finite perforated sample on which the affine temperature $u = x_j$ is imposed along the whole outer boundary, the hole walls being insulated, which is the setting of Eq. (14). In the bulk the solution is the two-scale expansion $x_j + \varepsilon\chi_j$, which however does not match the boundary data. The correction $\varepsilon\theta$ that restores it decays exponentially with the distance d to the boundary over a length proportional to ε, so that it is confined to a band of thickness $\delta \sim \varepsilon$ covering an area of about $4L\delta$, a fraction $4\delta/L = O(\varepsilon/L)$ of the sample.

Consequence for the conductivity tensor. Inserting the decomposition of Eq. (15) into the apparent tensor of Eq. (14) splits this average in two. The first part is the flux of $x_j + \varepsilon\chi_j(\{x/\varepsilon\})$, whose gradient $e_j + \nabla_y\chi_j$ is the same field in every cell. Its average over one cell is $\kappa^\star_{ij}$ by the definition of Eq. (11), and since the sample is made of a whole number of cells, its average over the sample is exactly $\kappa^\star_{ij}$. This is where the exact tiling matters, as partial cells along the edges would leave an additional error. What remains is the flux of the layer alone,

$$\kappa^{\mathrm{app}}_{ij} - \kappa^\star_{ij} = \frac{1}{|\Omega|}\int_{\Omega_{\mathrm{solid}}} \kappa\,\frac{\partial(\varepsilon\theta_j)}{\partial x_i}\,dx, \tag{17}$$

where θ_j is the correction of Eq. (15) for the direction j.

The two observations above are all that is needed to estimate this integral. By the first, the layer near a given edge can be written as $\varepsilon\theta_j(x) = \varepsilon\Theta_j(x/\varepsilon)$, with a profile Θ_j that does not depend on ε, and by the second this profile vanishes beyond about one cell. The chain rule gives $\partial_{x_i}(\varepsilon\theta_j) = \partial_{y_i}\Theta_j$ with $y = x/\varepsilon$, the factor ε in front cancelling the factor $1/\varepsilon$ produced by the differentiation. The gradient of the layer is therefore of order one, but it only lives within a distance of order ε from the boundary. Along the left edge, changing variables with $dx = \varepsilon^2\,dy$, the contribution of the layer to the integral is

$$\varepsilon^2 \int \kappa\,\partial_{y_i}\Theta_j\,dy = \varepsilon^2 \cdot \frac{L}{\varepsilon} \cdot c_{ij} = \varepsilon L\,c_{ij},$$

where L/ε is the number of cells along the edge and c_{ij} is the contribution of a single cell of the layer, which, like Θ_j, does not depend on ε. Each of the four edges contributes in this way, with its own constant, and dividing by $|\Omega| = L^2$ gives

$$\kappa^{\mathrm{app}}_{ij} - \kappa^\star_{ij} \approx C_{ij}\,\frac{4\varepsilon}{L}, \tag{18}$$

with C_{ij} the average of the four edge constants. In words, a flux error of order one is spread over a band of width $\delta \sim \varepsilon$ along the boundary, which covers an area of about $4L\delta$ and thus a fraction $4\delta/L = O(\varepsilon/L)$ of the sample. The only part of the layer this count misses are the four corners, each a patch of about

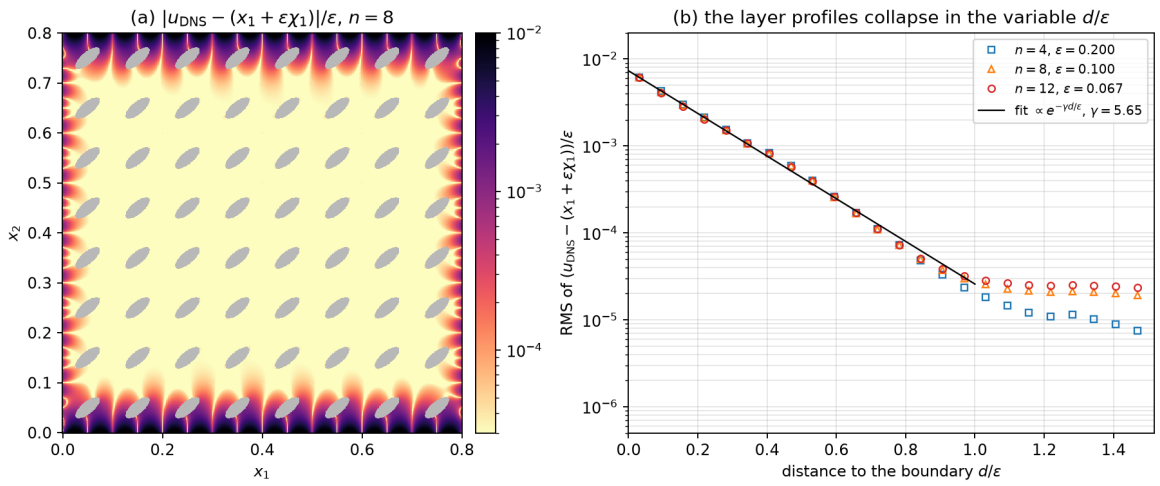


Figure 7: Measured boundary layer for the rotated ellipse ($a = 0.3$, $b = 0.12$, $40°$), with $u = x_1$ imposed on the whole outer boundary. Since the expansion $x_1 + \varepsilon\chi_1$ is exact in the bulk for this boundary data, its difference with the DNS is the boundary layer alone. (a) Map of this difference, divided by ε, for $n = 8$, on a logarithmic scale with the holes in grey. (b) Its RMS over strips of width $\varepsilon/16$ parallel to the boundary, against the distance to the boundary in units of ε, for three cell sizes. The three profiles coincide, which shows that the layer depends on x only through x/ε, and decay exponentially at the fitted rate $\gamma = 5.65$ until they reach the level of the discretization error a little under one cell from the boundary. Generated by `scripts/07_boundary_layer.py`.

one cell, which contribute $O(\varepsilon^2/L^2)$ and are one order smaller. Equation (18) predicts more than the rate $O(\varepsilon)$, since it says that the ratio of $\kappa^{\mathrm{app}} - \kappa^\star$ to $4\varepsilon/L$ should be the same number for every n, a sharper prediction that can be checked directly against the measured tensors. The constant itself is small for the holes studied here, because the layer starts from the small values of χ_j on the cell faces and decays within a fraction of a cell.

3.2.5 Results

Both comparisons with the DNS use the geometry and the elliptical hole of Section 3.2.3.

Field error: second-order accuracy of the reconstruction. The DNS field u_{DNS} is compared with two approximations, the macroscopic field T alone and the two-scale reconstruction $T + \varepsilon\chi \cdot \nabla T$ of Eq. (13), on a 40×40 grid of points covering $[0.15L, 0.85L]^2$, points falling inside a hole being skipped. Figure 8 shows the RMS of both differences against the cell size. The macroscopic field alone converges at first order, with a fitted rate

$$\mathrm{RMS}(u_{\mathrm{DNS}} - T) \sim \varepsilon^{0.97},$$

which is expected since it misses the oscillation of amplitude ε that the holes imprint on the temperature. Adding the corrector restores this oscillation, and the error of the reconstruction then decays at second order,

$$\mathrm{RMS}\big(u_{\mathrm{DNS}} - (T + \varepsilon\chi \cdot \nabla T)\big) \sim \varepsilon^{2.02}.$$

This is the central result of the study, and it is what the expansion of Eq. (13) predicts, the first neglected term being of the form $\varepsilon^2 \chi_{ij}(\{x/\varepsilon\})\, \partial_i \partial_j T$, with second-order correctors χ_{ij}, which does not vanish here because T is curved. The two rates open a gap that widens as the cells shrink, and at the smallest cell size the reconstruction is about two hundred times more accurate than T alone.

Two checks support this reading. The first concerns the finite element error. The cell and DNS meshes were refined in three levels, from $1.5/40$ to $1.5/120$ on the unit cell and from $\varepsilon/20$ to $\varepsilon/60$ in the DNS, and the open circles of Fig. 8 show that even the coarser levels lie close to the finest one. Between the last two levels the reconstruction error changes by at most one percent, so the plotted error is the homogenization error and not a discretization artefact. The second check concerns the two largest cells, $n = 2$ and 3, which are left out of the fit. The sampling region keeps a distance of $0.15L = 0.12$ from the edges, which is less than one cell for these sizes, so the sampled points reach into the boundary layers

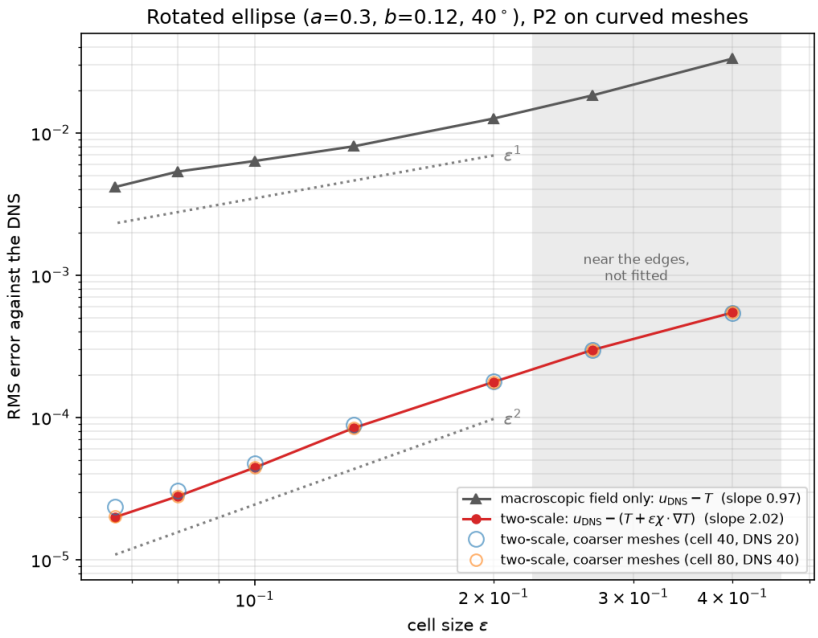


Figure 8: RMS difference between the DNS and the homogenized fields, as a function of the cell size ε, for the rotated ellipse ($a = 0.3$, $b = 0.12$, 40°), on a log-log scale. Triangles: macroscopic field T alone, first order. Filled circles: two-scale reconstruction $T + \varepsilon\chi \cdot \nabla T$, second order. Open circles: the same reconstruction error computed on the two coarser mesh levels; their agreement with the finest level shows that the FEM error is negligible. Slopes are fitted over $n \geq 4$; the shaded points ($n = 2, 3$) are affected by the boundary layers. Dotted lines: ε^1 and ε^2 references. This is the central quantitative result of the study. Generated by `scripts/04_homogenization_error.py`.

of Section 3.2.4, and with so few cells the expansion is in any case far from its asymptotic regime. The figure shows these two points departing from the second-order trend set by the smaller cells.

Effective conductivity: convergence of the full tensor. The second comparison concerns $\kappa^\star$ itself. An earlier version of this study compared $\kappa^\star_{11}$ with the effective conductivity read off the DNS of Eq. (3) for a circular hole, and found a relative difference that did not decrease with ε. In light of the discussion above this is no surprise, since for the circle the homogenized solution is exact, and the comparison could only measure the relative FEM error, which Section 2.3.3 showed to be constant. For an anisotropic hole the flux of Eq. (3) is moreover not $\kappa^\star_{11}$, because the insulated top and bottom edges interact with the off-diagonal coupling. We therefore compare $\kappa^\star$ with the apparent tensor of a finite sample defined in Eq. (14). As shown in Section 3.2.4, the two differ only through the flux of the boundary layer, by an amount predicted by Eq. (18) to be proportional to $4\varepsilon/L$, with a constant that does not depend on n.

Figure 9 shows the relative difference $\|\kappa^{\mathrm{app}} - \kappa^\star\|_F / \|\kappa^\star\|_F$ for the circle and the rotated ellipse, with $\kappa^\star$ computed on a fine unit-cell mesh. For both shapes it decreases at the fitted rate $\varepsilon^{1.00}$, with no sign of a plateau. The prediction of Eq. (18) holds in its sharper form as well. Dividing the relative difference by the area fraction $4\varepsilon/L$ of the layer gives the same number at every n, 4.28×10^{-3} for the circle and 6.3×10^{-4} for the ellipse, to within one percent. This holds even at $n = 2$, where the corners make up a large part of the layer but, being of higher order, still do not show.

The right panel of the figure follows the two diagonal entries of the ellipse separately. They converge at the same rate, and both differences are positive in every case, the finite sample always conducting slightly better than the effective material. The off-diagonal entry, which is not plotted, is reproduced about a thousand times more closely, at the level of the remaining FEM error, so that its rate cannot be resolved. Each DNS was also run at the two mesh resolutions $\varepsilon/30$ and $\varepsilon/60$, and given the h^4 convergence of these energy quantities, the difference between the two places the FEM error of the finer level at about one percent of the smallest difference measured.

The running times of the two approaches were not measured in this study. Comparing them on a fixed plate as the number of holes increases would show directly how much the homogenized pipeline

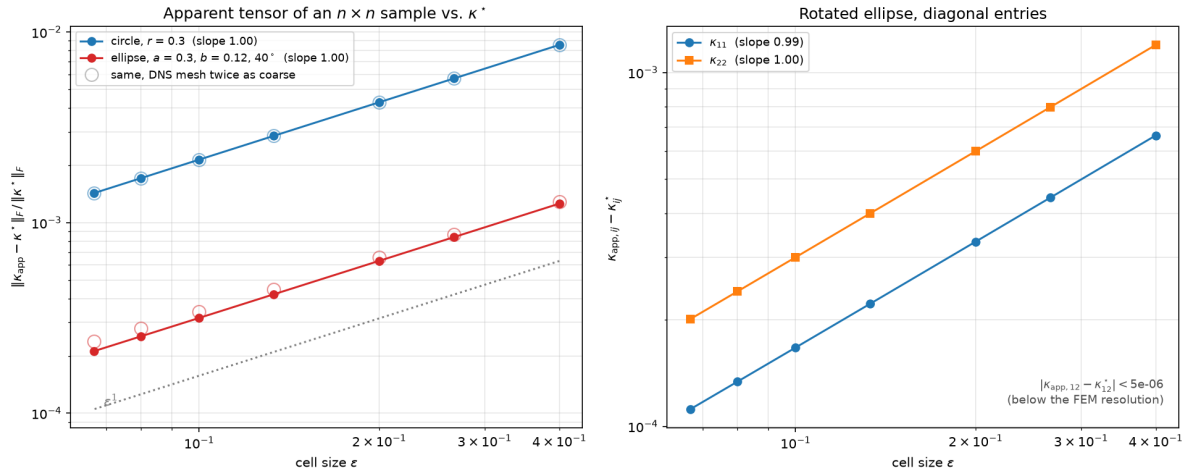


Figure 9: Left: relative difference between the apparent tensor κ^{app} of an $n \times n$ perforated sample (DNS with $u = x_j$ on the outer boundary, Eq. (14)) and the effective tensor $\kappa^\star$, as a function of the cell size, for a circle ($r = 0.3$) and the rotated ellipse ($a = 0.3$, $b = 0.12$, 40°). Filled symbols: DNS element size $\varepsilon/60$; open symbols: $\varepsilon/30$. Both decrease as ε^1, the width of the boundary layer. Right: the diagonal entries for the ellipse, $\kappa^{\text{app}}_{ii} - \kappa^\star_{ii}$, positive in every case; the off-diagonal entry differs by less than 5×10^{-6}. Generated by `scripts/05_tensor_convergence.py`.

saves, since its cost should not depend on ε, whereas the cost of the DNS grows quickly with the number of holes. Such a comparison would be simple to add to the existing scripts.

4 Conclusion

This study implemented a first-order periodic homogenization method for steady-state heat conduction in a plate perforated by insulating holes, validated it against a direct numerical solve, and used it to study how the shape of the holes determines the effective conductivity tensor. The direct solve served both as a benchmark and as a motivation, since its cost and the complexity of its geometry grow quickly with the number of holes. The homogenized model replaces it by a single unit-cell problem, from which the effective tensor $\kappa^\star$ is computed, followed by a solve on a plain mesh without holes.

The comparison with a direct solve on an exactly matching tiling confirms the two rates predicted by the two-scale expansion. For a rotated elliptical hole, the macroscopic field alone approximates the DNS to first order in ε, and the reconstruction that adds the corrector term $\varepsilon\chi \cdot \nabla T$ to second order, so that at the smallest cell size tested it is about two hundred times more accurate. The conductivity tensor of a finite perforated sample converges to $\kappa^\star$ at first order, from above, for both a circular and an elliptical hole. This rate is set by a boundary layer, which was measured to decay exponentially over a distance proportional to ε and therefore occupies a fraction $4\varepsilon/L$ of the sample, a prediction the measured tensors follow closely. None of this was visible before the discretization was controlled. With linear elements on meshes tied to ε, the finite element error does not shrink with ε and hides both rates behind a plateau, and a circular hole, for which the homogenized solution happens to be exact with the boundary conditions used, turned out to test the finite element error rather than the homogenization.

Varying the shape of the hole tunes the effective tensor in a way that follows its symmetries. At constant porosity, elongating an axis-aligned ellipse raises the conductivity along its major axis at the expense of its minor axis, and rotating it produces an off-diagonal entry, which records the sideways deflection of heat by the tilted hole, vanishes at the axis-aligned orientations and is largest near 45°.

Several extensions follow naturally. Measuring the running times of both approaches would make the efficiency argument quantitative. The code that tells hole walls apart from domain edges currently assumes smooth holes such as circles and ellipses, and a small change would extend it to polygonal holes such as triangles or squares, broadening the range of shapes that can be studied. The macroscopic domain could also be curved without difficulty, since the homogenized problem of Eq. (12) places no requirement on its shape. Finally, the method extends to three dimensions at the cost of a third corrector and larger meshes, and Figs. 10 and 11 in the appendix illustrate the geometry such a study would involve.

A Additional renderings

This appendix collects renderings, made with Gmsh and ParaView, that illustrate the study without being needed for its argument. The three-dimensional ones accompany the discussion of Section 2.2, and the plate solutions complement the single-cell fields of Fig. 4.

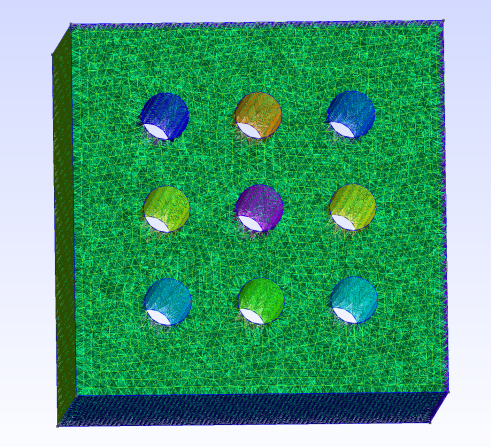


Figure 10: Mesh of a cube perforated by 3×3 parallel cylinders, opened in Gmsh, with first-order elements.

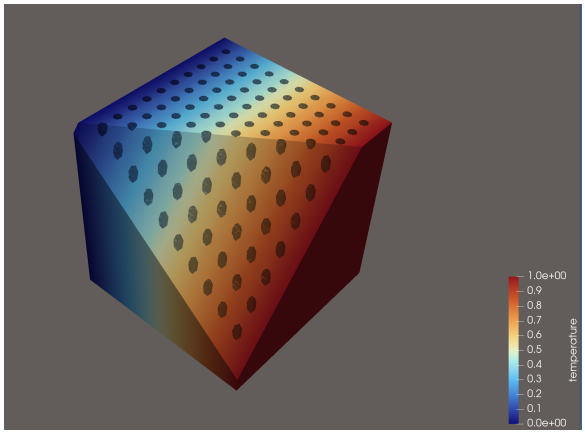


Figure 11: Temperature field on a cube perforated by 10×10 parallel cylinders, cut open to show its inside. The temperature varies across the cylinders but not along them.

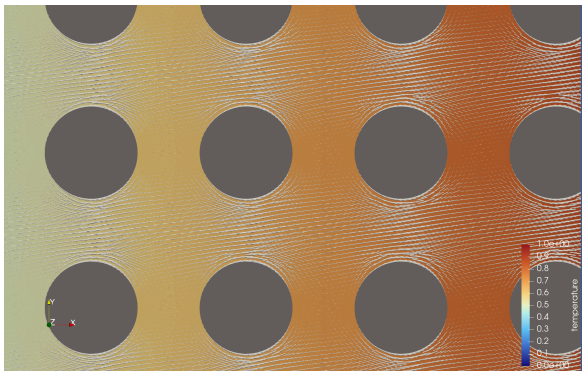


Figure 12: Temperature field and local heat flux, shown by arrows, over a whole plate perforated by circular holes. Parameters are $\kappa_{\text{solid}} = 1.0$, hole radius 0.3, 10 holes per side on the domain $L_1 = L_2 = 1.0$, so that $\varepsilon = L_1/10$, which give $\kappa^\star = \begin{pmatrix} 0.56 & \approx 0 \\ \approx 0 & 0.56 \end{pmatrix}$, isotropic and diagonal as expected for a circular hole.

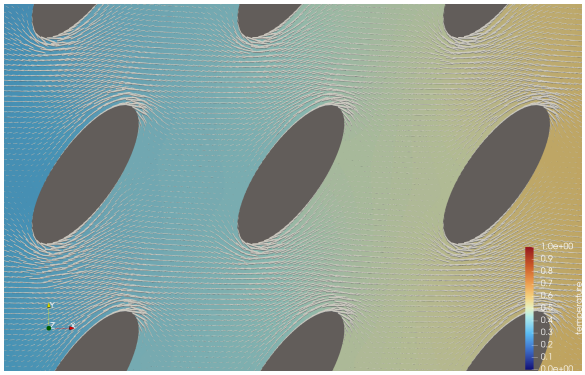


Figure 13: Temperature field and local heat flux, shown by arrows, over a whole plate perforated by elliptical holes. Parameters are $\kappa_{\text{solid}} = 1.0$, ellipse semi-axes $a = 0.4$ and $b = 0.15$ rotated by $55°$, 10 holes per side on the domain $L_1 = L_2 = 1.0$, so that $\varepsilon = L_1/10$, which give $\kappa^\star = \begin{pmatrix} 0.59 & 0.12 \\ 0.12 & 0.69 \end{pmatrix}$. The flux is largest near the tips of the major axis, where the heat is forced to curve most sharply around the hole.

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